

Mark Naeem

Robotics and Computer Vision Engineer



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Edinburgh, UK

A computer vision and robotics research and development engineer with demonstrated hands-on experience in industry and academia. A passionate self-learner with an urge to support open-source and research communities. A recognised team player in startups and prominent open-source projects, excelling in dynamic, self-directed, and challenging environments.

EXPERIENCE

Machine Learning Research Engineer

February 2024 – Present

Leap AI

Edinburgh, UK

- Researching and building foundation models to solve 3D spatial representation for robot manipulation (Embodied AI).
- Staying up-to-date with research trends to continuously prototype potential improvements.
- Planning software architecture for perception modules and the integration with the rest of other systems.

Robotics Software Engineer – Perception

January 2022 – January 2024

Kingdom Technologies Ltd

Glasgow, UK

Kingdom is an autonomous lawnmower manufacturer for large-scale fields. Working in a small team for a startup, I undertake a diverse range of responsibilities.

- Operating an extensive suite of sensors (LiDAR, stereo cameras, GNSS, IMU) in dynamic, unstructured environments shared with humans.
- Optimising and productising models for real-time, safety-critical tasks on resource-constrained devices.
- Working on state-of-the-art perception models for various tasks such as 3D object detection, point cloud segmentation, obstacle classification, and traversability estimation.
- Investigating different visual odometry techniques for GNSS-denied environments with limited visual features.
- Running CI/CD pipelines to streamline software deployment across a network of over 50 active assets.
- Navigated multiple software development lifecycles (SDLC) – from business requirement analysis and planning to the implementation of embedded drivers, high-level interfaces, the creation of unit/integration tests, and maintenance.
- Built MLOps pipelines for training, testing, deployment, and performance monitoring for various vision models.

Machine Learning Research Engineer

September 2020 – December 2021

Uniparticle

Cairo, Egypt

Uniparticle is a software house specialised in large-scale and government projects. I worked on various projects in different areas of machine learning such as machine vision, recommendation systems, and probabilistic modelling.

- Built IDeepify, a complete end-to-end deep learning-based KYC (Know Your Customer) pipeline.
- Published a novel Hidden Markov Model-based technique to analyse assessment results in collaboration with Discovery Education. The technique was validated with a real-life dataset.
- Built BKT-CAT, a Bayesian knowledge tracing computerised adaptive testing system.
- Used knowledge space theory and recommendation systems along with adaptive testing results to build an adaptive learning engine.
- Introduced various probabilistic student modelling and simulation techniques to improve adaptive quizzes quality.
- Improved the performance of an existing adaptive testing simulation algorithm, optimising it to run 20 times faster.

Teaching Assistant - Machine Vision

September 2020 – February 2021

Ain Shams University

Cairo, Egypt

- Tutored and prepared the material for the machine vision course.
- The course covered traditional computer vision techniques and modern machine learning-based approaches.

PUBLICATIONS

Bayesian Knowledge Tracing for Assessment Results Analysis

February 2022

IEEE

[Link to the paper](#)

Bayesian Knowledge Tracing, a Hidden Markov model, is utilised to analyse assessment results. The technique was tested and validated with a real-life dataset. This was a collaboration research project between Discovery Education and Uniparticle.

Linear Time Invariant State Space System Identification Using Adam Optimization

February 2020

IEEE

[Link to the paper](#)

A new system identification method is proposed. A state space model is presented with a TensorFlow graph. Adam optimisation is used to optimise the learnable state and input matrices.

PROJECTS

VSLAM Playground

A full Visual SLAM pipeline using deep learning-based feature descriptors and matchers and deep stereo depth estimation.

Depth Yolact ROS

A real-time 3D object detection and pointcloud instance segmentation for RGBD images.

IDeepify

Robust deep learning-based KYC product consisting of face recognition and matching, text segmentation, data extraction, and OCR. The model was trained with mostly synthetic data and a few real-life labelled examples.

Swerve Steering Controller – ROS Controller Package

A real-time safe N-wheel non-holonomic omnidirectional vehicle controller.

Deep Computer-Aided Sperm Analysis (CASA)

Faster R-CNN object detection and a modified DeepSORT tracking were used to detect and track human spermatozoa in phase-contrast, dark-field, and bright-field microscopy imaging.

Move Base Sequence – ROS Package

A ROS action server to handle multiple goals and track execution with ROS navigation stack.

D435i Visual-Inertial Odometry and SLAM

D435i camera is solely used to obtain reliable visual-inertial odometry and SLAM.

AWARDS

- Al-Alfi Foundation Scholarship for outstanding undergraduate students
- Erasmus+ research grant at the University of Central Lancashire (UCLan)
- Graduation project grant organised by the University of Lincoln and Ain Shams University
- Best project of the year for design of mechatronic systems course
- 1st place in FSUK18 C&M, honoured by the Egyptian president in the National Youth Conference 2018, Egypt

SKILLS

C/C++, Python, Git, CI/CD, Linux, RTOS, Embedded Software, Docker, Kubernetes, MATLAB, ROS/ROS2, ADAS, State Estimation, Localisation, SLAM, Visual Odometry, Sensor Fusion, Multi-Sensor Calibration, Motion Planning, CUDA, Reinforcement Learning, TensorFlow, PyTorch, Teamwork, Leadership, Self-Discipline, Problem-Solving

EDUCATION

Ain Shams University

September 2015– June 2020

B.Sc. in Mechatronics Engineering – Robotics - Class of 2020

Five-year degree

- **GPA:** 3.86/4.00
- Ranked Top 5 (3rd)
- **Graduation Project:** Autonomous mobile manipulator for fruit picking – A collaboration between Ain Shams University and University of Lincoln. (project GPA: 4.00/4.00 A+)
- **Activities:** ASU Racing Team FSUK'18, ASU-Sussex racing team FSUK'19, Erasmus+ Mobility Project